

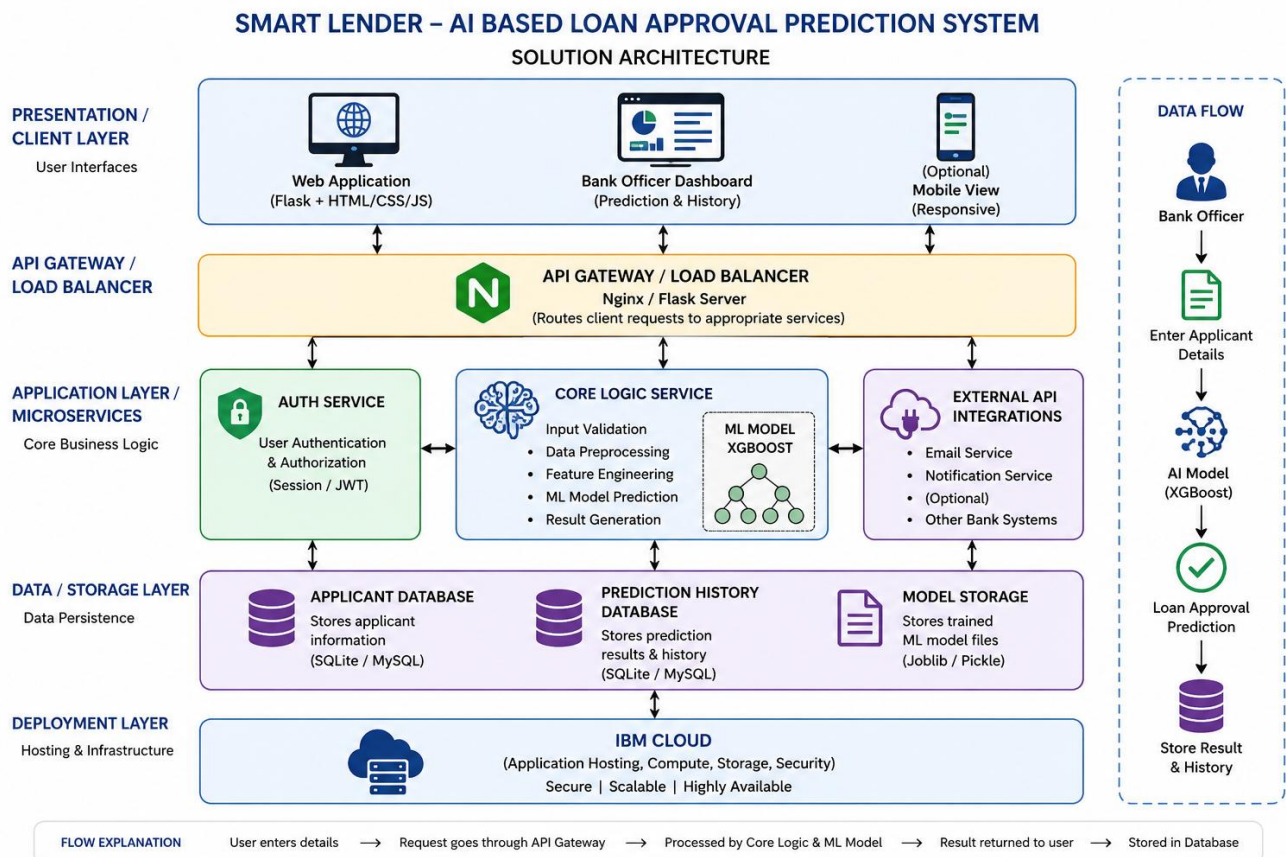
Solution Architecture

Date	29 June 2026
Team ID	SWTID-2026-7290
Project Name	Smart Lender
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.

Solution Architecture Diagram



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Allows users to enter applicant details and view prediction results through a responsive interface.	HTML5, CSS3, Bootstrap, JavaScript
API Gateway / Server	Receives requests, validates input, and forwards data to the prediction service.	Flask
Core Logic Service	Performs preprocessing and executes the trained XGBoost model to generate loan approval predictions.	Python, Scikit-learn, XGBoost, Pandas, NumPy
Database	Stores applicant information, prediction history, and user data.	SQLite (or MySQL)
Cloud Deployment	Hosts the application and enables online access.	IBM Cloud